Cisco PE Network Device Routing Protocol Flap Failure SOP for KR-PE-15 (KR Puram Access)

Node: KR-PE-15 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_4149

Description

This Standard Operating Procedure (SOP) is designed to address the failure domain related to routing protocol flap on the Cisco PE device KR-PE-15 at the KR Puram Access site. The routing protocol flap failure is primarily attributed to transient issues within the BGP (Border Gateway Protocol) and OSPF (Open Shortest Path First) routing protocols, culminating in ISIS (Intermediate System to Intermediate System) LSP (Label Switched Path) flooding. The PE device, being a critical component of the network architecture, plays a pivotal role in interconnecting the access and core networks. The Cisco IOS software, running on the PE device, employs the BGP and OSPF protocols to facilitate dynamic routing, while the ISIS protocol is used for intra-domain routing. During periods of network instability, these protocols can exhibit transient issues, leading to routing protocol flap events. When the BGP route churn, OSPF adjacency loss, and ISIS LSP flooding occur, the ALARM_4149 is triggered, indicating a warning-level alert.

Pre-requisite Checks

- Verify the physical connectivity of the device and its redundant links.

- Ensure that the remote hands are dispatched to the KR Puram Access site to facilitate access to the device.

- Conduct a thorough review of the device's configuration and logs to identify potential causes of the routing protocol flap.

Potential Root Causes

Cause	Probability
BGP route churn due to neighbor relationship issues	High
OSPF adjacency loss due to network topology changes	Medium
ISIS LSP flooding due to misconfiguration	Low

Diagnostic Commands & Expected Outputs

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Command: show ip bgp summary, Expected Output: BGP table version is 1, main routing table version 1 BGP table version 1, neighbor version 1 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State PrefRcvd, Description: This command generates a summary of the BGP table, displaying the major statistics of each BGP peer, facilitating the identification of any issues with the neighbor relationship.

Command: show ip ospf neighbor, Expected Output: OSPF Neighbor Status for Process ID 1 Neighbor ID Pri State Dead Time Address Interface,
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Description: This command displays a list of all OSPF neighbors, providing information on the neighbor's state, dead time, address, and interface, enabling the identification of OSPF adjacency loss issues.

Command: show isis database, Expected Output: IS-IS Level-1 Link-State Database: LS age: 0, Sequence number: 0x3e8 LSP ID: 192.168.1.1.00-00-00, Sequence number: 0x3e8 Checksum: 0x4a5f Length: 24 IS-IS Level-2 Link-State Database: LS age: 0, Sequence number: 0x3e8 LSP ID: 192.168.2.2.00-00-00, Sequence number: 0x3e8 Checksum: 0x4a5f Length: 24, Description: This command displays the ISIS LSP database, providing information on the LSP IDs, sequence numbers, checksums, and lengths, enabling the identification of ISIS LSP flooding issues.
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Resolution Steps

Step 1: Step: Reset the BGP neighbor relationship, Command: clear ip bgp *, Description: This command resets the BGP neighbor relationship, clearing the BGP table and re-establishing the connection to the neighbor.

Step 2: Step: Re-establish the OSPF adjacency, Command: clear ip ospf process, Description: This command re-establishes the OSPF adjacency, re-building the OSPF neighbor relationship and re-sending OSPF hello packets.

Step 3: Step: Clear the ISIS LSP flooding, Command: clear isis database, Description: This command clears the ISIS LSP flooding, re-building the ISIS LSP database and re-sending ISIS hello packets.

Post-Check Verification

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Command: show ip bgp summary, Expected Output: BGP table version is 1, main routing table version 1 BGP table version 1, neighbor version 1 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State PrefRcvd, Description: This command verifies that the BGP table has been re-built and the neighbor relationship has been re-established.

Command: show ip ospf neighbor, Expected Output: OSPF Neighbor Status for Process ID 1 Neighbor ID Pri State Dead Time Address Interface, Description: This command verifies that the OSPF adjacency has been re-established and the OSPF neighbor relationship has been re-built.

Command: show isis database, Expected Output: IS-IS Level-1 Link-State Database: LS age: 0, Sequence number: 0x3e8 LSP ID: 192.168.1.1.00-00-00, Sequence number: 0x3e8 Checksum: 0x4a5f Length: 24 IS-IS Level-2 Link-State Database: LS age: 0, Sequence number: 0x3e8 LSP ID: 192.168.2.2.00-00-00, Sequence number: 0x3e8 Checksum: 0x4a5f Length: 24, Description: This command verifies that the ISIS LSP flooding has been cleared and the ISIS LSP database has been re-built.
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Roll-back Procedures

- Step: Restore the original BGP configuration, Command: copy running-config startup-config, Description: This command restores the original BGP configuration, re-establishing the original BGP neighbor relationship.

- Step: Restore the original OSPF configuration, Command: copy running-config startup-config, Description: This command restores the original OSPF configuration, re-establishing the original OSPF neighbor relationship.

- Step: Restore the original ISIS configuration, Command: copy running-config startup-config, Description: This command restores the original ISIS configuration, re-establishing the original ISIS LSP flooding.